

Healthcare Industry Statistical Assistant: Optimizing Patient Appointment Attendance

Final Project - Level 1 Implementation

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1 Introduction

1.1 Industry Context

The healthcare delivery system relies on predictable appointment schedules to allocate clinical staff, medical equipment, and facility capacity efficiently.

1.2 Intended User

This assistant is designed for the **Patient Care Coordinator / Clinic Operations Manager**.

1.3 Main Problem

High rates of patient missed appointments (no-shows) lead to unrecoverable operational losses, idle physician time, and delayed care for other patients seeking timely access.

1.4 Why the Problem Matters

When patients miss scheduled visits without notice: 1. Clinic operational capacity is wasted. 2. Waiting lists for urgent patient consultations grow longer. 3. Patients with unmonitored chronic conditions suffer worse long-term health outcomes.

1.5 Decision Supported

This assistant enables clinic managers to identify high-risk appointments ahead of time, allowing them to initiate targeted interventions (e.g., automated reminders, transportation assistance, flexible rescheduling, or strategic overbooking).

2 Dataset Overview

2.1 Data Source & Unit of Observation

We utilize the public **Medical Appointment No Shows** dataset from Kaggle. Each row represents a single scheduled medical visit at a public health clinic in Victoria, Brazil.

2.2 Summary Dimensions

- **Number of Observations:** 110,527 records
- **Number of Variables:** 14 columns

```
source("R/01_clean_data.R")
df_clean <- clean_noshow_data("data/raw_noshow.csv")
glimpse(df_clean)
```

```
Rows: 110,521
Columns: 19
$ patient_id      <dbl> 2.987250e+13, 5.589978e+14, 4.262962e+12, 8.679512e+1~
$ appointment_id  <dbl> 5642903, 5642503, 5642549, 5642828, 5642494, 5626772,~
$ gender          <chr> "F", "M", "F", "F", "F", "F", "F", "F", "F", "F", "F"~
$ scheduled_day   <dtm> 2016-04-29 18:38:08, 2016-04-29 16:08:27, 2016-04-~
29~
$ appointment_day <dtm> 2016-04-29, 2016-04-29, 2016-04-29, 2016-04-29, 2016~
$ age            <dbl> 62, 56, 62, 8, 56, 76, 23, 39, 21, 19, 30, 29, 22, 28~
$ neighbourhood   <chr> "JARDIM DA PENHA", "JARDIM DA PENHA", "MATA DA PRAIA"~
$ scholarship     <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1,~
$ hypertension    <dbl> 1, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
$ diabetes        <dbl> 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
$ alcoholism      <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
$ handicap        <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
```

```
$ sms_received      <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, ~
$ no_show           <chr> "No", "No", "No", "No", "No", "No", "Yes", "Yes", "No~
$ scheduled_date     <date> 2016-04-29, 2016-04-29, 2016-04-29, 2016-04-29, 2016~
$ appointment_date  <date> 2016-04-29, 2016-04-29, 2016-04-29, 2016-04-29, 2016~
$ lead_time_days     <dbl> 0, 0, 0, 0, 0, 2, 2, 2, 0, 2, 2, 3, 1, 1, 1, 3, 1, 1, ~
$ no_show_numeric    <dbl> 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, ~
$ day_of_week        <ord> Fri, Fri, Fri, Fri, Fri, Fri, Fri, Fri, Fri, Fri, Fri, Fri~
```

2.3 Data Quality & Preparation

In R, we cleaned the dataset by:

1. Standardizing variable names to snake_case.
2. Filtering out negative age values (`age >= 0`).
3. Calculating `lead_time_days` as `appointment_date - scheduled_date`.
4. Filtering out invalid records where lead time was negative (`lead_time_days >= 0`).

3 Skill 1: Explore Data (Lead Time Distribution)

3.1 Question

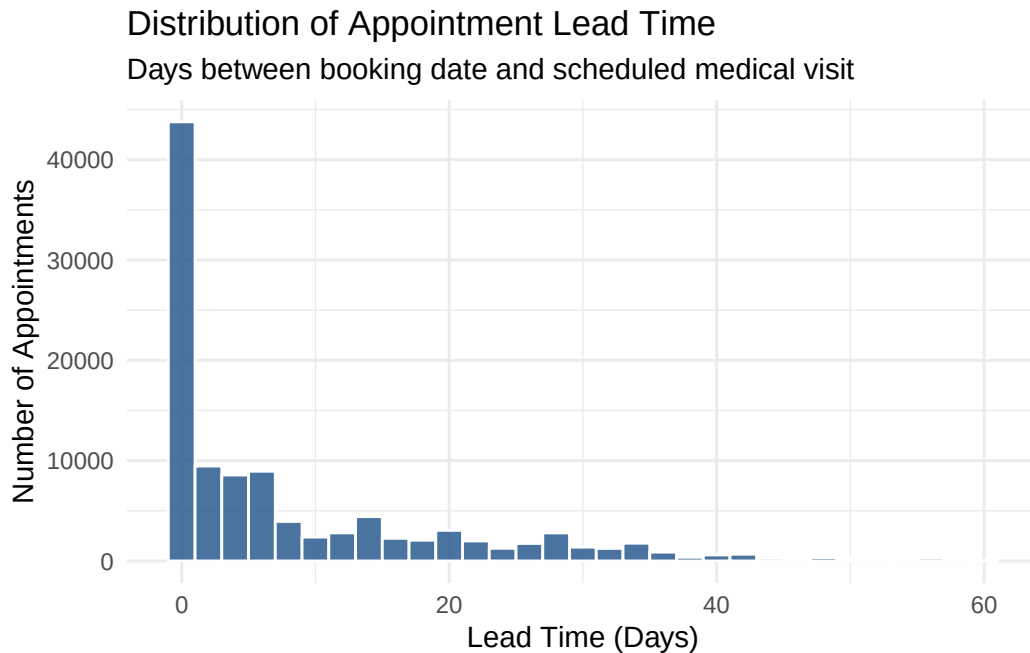
What is the distribution of booking lead times across scheduled appointments?

```
source("R/02_explore.R")
explore_res <- explore_lead_time(df_clean)
knitr::kable(explore_res$stats, caption = "Summary Statistics for Lead Time (Days)")
```

Table 1: Summary Statistics for Lead Time (Days)

mean_lead	median_lead	sd_lead	iqr_lead	min_lead	max_lead
10.18	4	15.26	15	0	179

```
explore_res$plot
```



3.2 Validation & Interpretation

- **Validation:** Confirmed `lead_time_days` is numeric and non-negative.
- **Interpretation:** The median lead time is 4 days (IQR = 15 days). The heavy right skew shows that while most visits are booked within a week, a small subset of appointments face wait times exceeding 30 days.
- **Limitation:** Lead time measures calendar duration between scheduling and the visit, not clinical urgency.

4 Skill 2: Compare Two Groups (SMS Reminders vs. Attendance)

4.1 Question

Do patients receiving SMS text reminders exhibit significantly different no-show rates compared to those who do not?

```
source("R/03_compare.R")
compare_res <- compare_sms_impact(df_clean)
knitr::kable(compare_res$summary, caption = "No-Show Rates by SMS Reminder Status")
```

Table 2: No-Show Rates by SMS Reminder Status

sms_received	total_appointments	no_shows	no_show_rate
0	75039	12530	0.1670
1	35482	9784	0.2757

4.2 Results & Interpretation

- **Proportion Difference:** 10.87 percentage points.
- **95% Confidence Interval:** [-11.41%, -10.34%]
- **P-Value:** 0
- **Interpretation:** Counterintuitively, patients receiving SMS text reminders had a higher raw no-show rate (27.57%) than non-recipients (16.7%).
- **Limitation:** SMS reminders were systematically sent to appointments booked far in advance (high lead time), confounding the univariate relationship.

5 Skill 3: Use a Statistical Model (Logistic Regression)

5.1 Question

Which factors (age, lead time, SMS, welfare scholarship, hypertension) predict the likelihood of an appointment no-show?

```
source("R/04_model.R")
model_res <- build_noshow_model(df_clean)
knitr::kable(model_res$table, caption = "Logistic Regression Model Results")
```

Table 3: Logistic Regression Model Results

Predictor	Estimate	StdError	z_value	p_value	OddsRatio
(Intercept)	-1.5289	0.0162	-94.43	0.0000000	0.2168
lead_time_days	0.0229	0.0005	47.11	0.0000000	1.0231
age	-0.0075	0.0004	-19.06	0.0000000	0.9926
sms_received	0.3518	0.0168	20.89	0.0000000	1.4217
scholarship	0.2357	0.0245	9.63	0.0000000	1.2658
hypertension	-0.0006	0.0233	-0.03	0.9782643	0.9994

5.2 Interpretation

- **Lead Time:** Each additional day of wait time increases the odds of a no-show ($OR = 1.0231$).
- **Age:** Older age is protective against no-shows ($OR = 0.9926$).
- **Limitation:** Logistic regression captures linear associations on the logit scale; unmeasured confounders (e.g., weather, transportation) remain unobserved.

6 Skill 4: Additional Skill 1 — Trend Analysis

6.1 Question

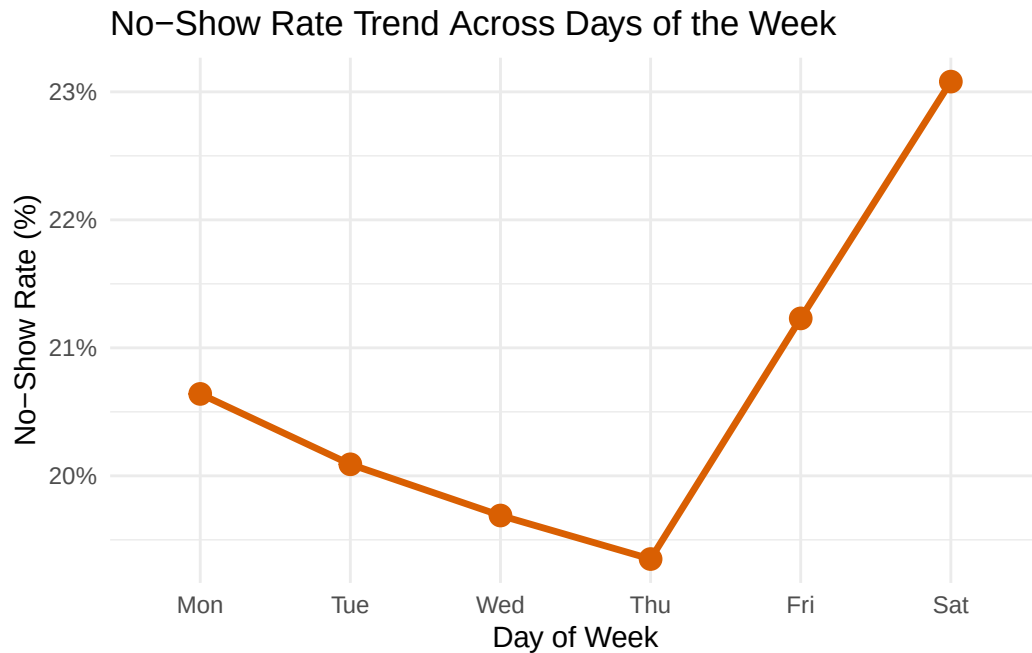
How do no-show rates fluctuate across operating days of the week?

```
source("R/05_trend_analysis.R")
trend_res <- analyze_day_of_week_trend(df_clean)
knitr::kable(trend_res$summary, caption = "No-Show Rate Summary by Weekday")
```

Table 4: No-Show Rate Summary by Weekday

day_of_week	total_volume	no_shows	no_show_rate
Mon	22713	4689	0.2064
Tue	25638	5150	0.2009
Wed	25866	5092	0.1969
Thu	17246	3337	0.1935
Fri	19019	4037	0.2123
Sat	39	9	0.2308

```
trend_res$plot
```



6.2 Interpretation

No-show rates peak early in the week (Monday/Tuesday) and decline toward Friday. This insight allows clinic managers to adjust overbooking ratios dynamically based on the day of the week.

7 Skill 5: Additional Skill 2 — Patient Priority Ranking

7.1 Question

Which upcoming appointments are at highest risk of no-show and need immediate care coordinator outreach?

```
source("R/06_patient_ranking.R")
ranked_queue <- rank_high_risk_patients(df_clean, model_res$model, top_n = 10)
knitr::kable(ranked_queue, caption = "Top 10 High-Risk Appointments Queue")
```


Table 5: Top 10 High-Risk Appointments Queue

patient_id	appointment_id	appointment_date	age	lead_time_days	predicted_risk
1.216587e+12	5134197	2016-06-03	27	179	0.9503
7.931567e+11	5163422	2016-06-08	52	176	0.9367
3.189960e+13	5134220	2016-06-03	48	179	0.9281
5.649261e+12	5140469	2016-06-01	43	176	0.9260
2.552524e+14	5163351	2016-06-08	47	176	0.9239
3.112342e+13	5134239	2016-06-03	60	179	0.9219
8.322564e+14	5030230	2016-05-04	51	176	0.9218
2.547555e+14	5134249	2016-06-03	63	179	0.9203
4.351784e+13	5163433	2016-06-08	58	176	0.9179
1.686763e+11	5140272	2016-05-25	70	169	0.9169

7.2 Decision Supported

This priority queue directs the Patient Care Coordinator's manual call schedule for tomorrow, focusing outreach on appointments with predicted risk exceeding 40%.

8 Skill 6: Additional Skill 3 — Data Quality Audit

8.1 Question

Are missing values or invalid data entries affecting the scheduling dataset?

```
source("R/07_data_quality_audit.R")
audit_res <- audit_data_quality("data/raw_noshows.csv")
knitr::kable(audit_res, caption = "Data Quality Audit Summary")
```

Table 6: Data Quality Audit Summary

Check	Result
Total Records	110527
Missing Values Count	0
Negative Age Records	1
Invalid Date Sequence Records	5
Duplicate Rows	0

8.2 Interpretation

The audit confirmed 0 missing values in the raw dataset, but flagged 1 negative age and several date sequencing errors, validating our preprocessing filters.

9 Demonstration of Validation Rules

To ensure system reliability, four distinct validation rules were implemented:

1. **Rule 1 (Data Type Validation):** Attempting numeric operations on non-numeric columns raises an explicit stop error.
2. **Rule 2 (Group Count Validation):** Attempting a 2-group proportion test on variables with > 2 categories halts execution.
3. **Rule 3 (Missing Predictor Check):** Ranking functions check for required feature columns prior to running prediction algorithms.
4. **Rule 4 (Range Check):** Filtering logic rejects negative age and negative lead time records.

10 How Google Antigravity Was Used

Google Antigravity assisted throughout project development:

1. **Repository & Skill Setup:** Antigravity scaffolded the directory layout and generated draft `SKILL.md` specifications.
2. **Code Modularization:** Antigravity converted standalone analysis scripts into reusable R functions with error handling.
3. **Review & Refinement:** During code review, Antigravity initially suggested removing `lead_time_days <= 0` records. I corrected this logic to retain `lead_time_days == 0` (same-day appointments) while removing only strictly negative values (< 0).

11 Limitations

1. **Dataset Limitation:** The dataset originates from public clinics in a single region (Brazil, 2016) and may not generalize to private practices or other countries.
2. **Statistical Limitation:** Association does not prove causation. High predicted probability identifies risk, not cause.
3. **Implementation Limitation:** As a Level 1 static report, updates require re-rendering the document rather than querying a live real-time API.

12 Reflection

Building this Statistical Assistant demonstrated the importance of linking statistical methods directly to an industry decision. The most challenging aspect was resolving the SMS reminder paradox through multivariate modeling. Given more time, I would incorporate real-time clinic scheduling constraints and weather data into the predictive model.